Gradient of a line

1 Write an expression for the gradient of the line that is inclined at an angle of 33° to the positive x-axis.

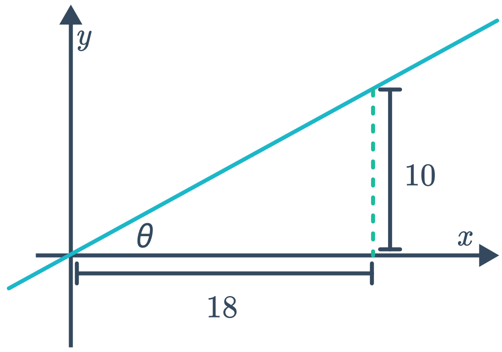
2 Find the gradient of a line that is inclined at an angle of 50° to the positive x-axis . Round your answer to one decimal place.

3 Find the angle of inclination of a line to the positive x-axis, whose gradient is 4. Round your answer to the nearest degree.

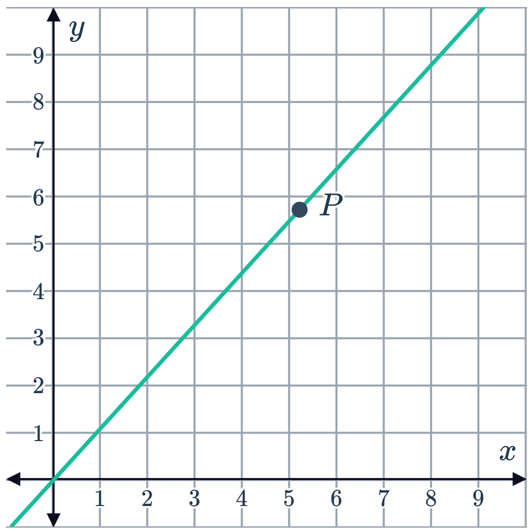
4 Consider the given line with an angle of inclination of θ with the positive x-axis:

a Find the gradient of the line.

b Hence, find θ. Round your answer to two decimal places.



5 Point $P\,(5.2, 5.72)$ lies on the given line passing through the origin:
Find θ, the angle of inclination of the line with the positive x-axis. Round your answer to two decimal places.

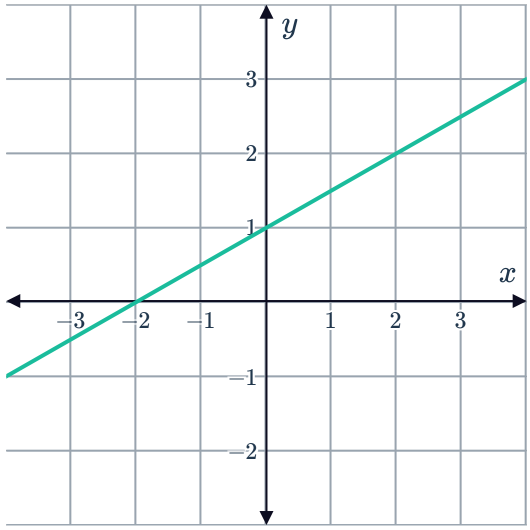


6 For each of the linear equations below, find θ the angle of inclination of the line with the positive x-axis. Round your answer to the nearest degree.

a $y = 6x + 2$

b $5x + 6y - 2 = 0$

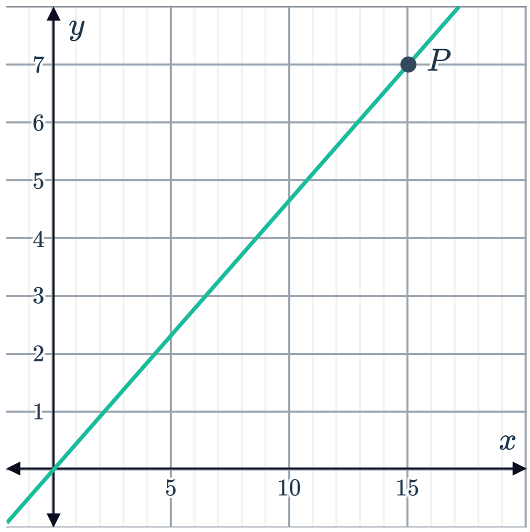
7 Consider the graph of the line $y = \frac{1}{2}x + 1$ given and the triangle whose vertices are the origin, the x-intercept and the y-intercept. θ is the acute angle in the triangle that the line makes with the x-axis.



a State the gradient of the line.

b Find the value of $\tan\theta$.

8 Point $P(15, 7)$ lies on the given line passing through the origin:
Find θ, the angle of inclination of the line with the positive x-axis. Round your answer to two decimal places.



9 A line makes an angle of θ with the positive x-axis. If $\theta < 90^\circ$. State whether the gradient of the line is positive or negative.

10 If a line has a gradient of -4 and an angle of inclination of θ with the positive x-axis, which of the following is true of the value of θ?

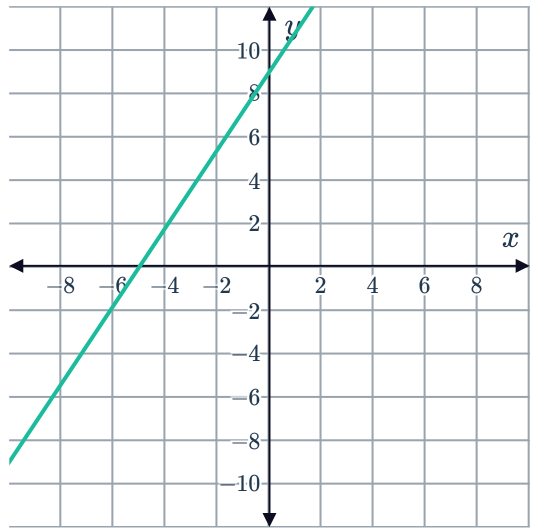
A $-90^\circ < \theta < 0^\circ$ **B** $0^\circ < \theta < 90^\circ$ **C** $90^\circ < \theta < 180^\circ$ **D** $180^\circ < \theta < 270^\circ$

11 Two points $A(1, -2)$ and $B(9, 30)$ lie on a line that makes an angle of θ degrees with the positive x-axis.

a Find the gradient of the line.

b Find θ to two decimal places.

12 Find θ, the angle of inclination of the line with the positive x-axis, if the x-intercept is -5 and the y-intercept is 9. Round your answer to two decimal places.



13 A line with gradient m has an angle of inclination of 65° with the positive x-axis, such that $\tan 65^\circ = m$. Without solving for m, find the value of θ such that $\tan\theta = -\dfrac{1}{m}$.

14 Find θ, the angle of inclination of the line with the positive x-axis for a line with a gradient of $-\dfrac{1}{\sqrt{3}}$.

15 A horizontal line makes an angle of 0° with the x-axis.

- **a** Evaluate $\tan 0^\circ$ by considering its relationship to the gradient of the line.
- **b** A vertical line is perpendicular to a horizontal line. Using the fact that the gradients of perpendicular lines have the relationship $m_1 \times m_2 = -1$, find the value of $\tan 90^\circ$.

16 A line is drawn on the cartesian plane at an angle of inclination θ with the positive x-axis. Given that $\sin\theta = \dfrac{12}{13}$, find the gradient m of the line.

Applications

17 Line L_1 makes an angle of θ with the positive x-axis. Line L_2 is such that it is perpendicular to line L_1. Form an expression for the gradient of line L_2 in terms of θ.

18 A line passing through the points $(2, -3)$ and $(x, 9)$ makes an angle of 120° with the positive x-axis. Find the exact value of x.

19 The steepness of a ski run is defined by the angle of inclination of the run with the horizontal ground.
By how many degrees is Run A steeper than Run B? Round your answer to one decimal place.